overlaps the next one over a length 2ε, and the ε-neighborhood of each point in the overlapping section will certainly be part of at least one of the two adjacent cells (for instance, point p in the figure has its ε-neighborhood completely contained in S_1). The trick is that, therefore, if p is a core point in one of the shards, it will also be a core point in the union of the shards, and hence its neighbors on both sides of the cells' border should be directly-reachable[17] from p, and in turn end up in the same cluster: therefore, if p is assigned to cluster C_1 for shard S_1 and to cluster C_2 for shard S_2, it follows that clusters C_1 and C_2 should be merged.

Vice versa, if we consider any point that is outside the shard's border, like r in figure 13.12, that's on the left of S_1's inner margin, then we know for sure that:

- It's distance from the shard's border is larger than ε, and therefore it's ε-neighborhood doesn't intersect S_1's outer margin;
- If there is a point z in S_2 that is reachable from r, then there must be a chain core points that are directly-reachable from z and r (for the definition of reachability – see section 12.3.1) and at least one of these points, call it w, must be in either S_1's inner or outer margin (because these areas extend exactly for a length equal to 2ε, which is exactly the diameter of core points' ε-neighborhood).
- Therefore, we can ignore r, because we'll join its cluster to z's when we examine point w.